

Quantifying Tail Risks in Brent Crude Oil Prices under Geopolitical Uncertainty: A Growth-at-Risk Approach

Abstract

Crude oil prices act as a key macroeconomic variable, often serving as a primary input in global inflation dynamics, industrial production costs, and international trade balances. The propagation of geopolitical tensions through energy markets can generate heightened volatility, which presents significant macroeconomic consequences and potential systemic risks for policymakers, firms, and institutional investors.

Many traditional econometric approaches assume that the average historical effect of a geopolitical shock adequately represents the underlying stochastic process. However, in the context of crude oil prices and geopolitical shocks—where data tend to exhibit substantial heteroskedasticity, asymmetric responses, and influential tail observations that can distort linear estimates—this assumption may prove overly restrictive and potentially misleading.

To help address the limitations of conditional mean forecasting, the present research adapts the Growth-at-Risk framework to model the broader conditional distribution of future crude oil price growth. Rather than focusing solely on average baseline effects, the analysis estimates how fluctuations in geopolitical risk might shift specific quantiles of the return distribution. This approach aims to capture heterogeneity across outcomes and estimate both downside and upside market vulnerabilities more effectively. The empirical strategy relies on the European Brent crude oil spot price and dictionary-based Geopolitical Risk Indices. Drawing on daily returns and a robust set of control variables, including speculative volatility proxies and exchange rate dynamics, the study specifies multiple quantile regressions to better isolate the tail-risk impact of geopolitical uncertainty.

Keywords: Geopolitical Risk, Brent Crude Oil, Quantile Regression, Growth-at-Risk, Tail Risk, Volatility.

JEL Codes: C32, F50, Q43.

1 Introduction

The study of geopolitical risk and its cascading effects on crude oil prices remains a significant concern for the global economy. While inflation dynamics vary due to a multitude of complex macroeconomic factors, oil prices serve as one of their primary drivers, often influencing the cost of global industrial production and affecting international trade balances and current account deficits. Geopolitical shocks—ranging from localized military conflicts and diplomatic standoffs to sweeping international embargos and maritime blockades—tend to propagate rapidly through energy markets. These events can introduce a level of profound uncertainty that may prompt immediate adjustments in consumption, investment, and precautionary storage among economic agents. Understanding not only whether geopolitical risk generally affects oil prices, but how vulnerable future prices might be to the extreme tails of these global risks, is therefore a highly valuable endeavor for modern economic stabilization and financial risk management.

Standard macroeconomic and econometric frameworks have historically tended to favor modeling the conditional mean of asset prices and macroeconomic indicators. By frequently relying on Ordinary Least Squares regression techniques, classical studies often operate under the assumption that the average historical effect of a geopolitical shock represents the most likely future outcome, treating variance as a relatively constant dispersion around that mean. However, the observable dynamics of global oil markets frequently challenge this assumption. Energy market data commonly exhibit intense volatility clustering, asymmetric shock responses, and heavy-tailed distributions. When financial data contain such pronounced heteroskedasticity and influential tail observations, linear conditional mean estimates can become systematically distorted. The average effect may be skewed by extremes without adequately capturing them, which can render standard policy guidance based solely on the mean overly restrictive and potentially misleading.

Consequently, the current research paradigm increasingly broadens its focus beyond average effects to emphasize tail risk exposure. Tail risk exposure often represents a highly relevant metric for economic agents operating under deep uncertainty. Policymakers can benefit from detailed assessments of downside risks to help design monetary and fiscal stabilization policies, particularly in environments where inflation targeting is a priority. Concurrently, energy firms, refineries, and institutional investors generally need to evaluate the potential for extreme upward or downward price movements to better manage their exposure, allocate capital, and hedge portfolios effectively. By explicitly modeling the tails of the conditional distribution, this research aims to provide a more granular, realistic, and policy-relevant characterization of oil price dynamics under geopolitical duress.

2 Background (Literature Review)

The methodological foundation of this research builds upon the Vulnerable Growth paradigm introduced by Adrian, Boyarchenko, and Giannone (2019). By modeling the conditional distribution of gross domestic product growth as a function of economic and financial conditions, their framework observed that deteriorating financial conditions tend to increase conditional volatility and shift the lower quantiles downward, while upper quantiles remain largely stable. This paradigm demonstrated that downside risks exhibit significant variance based on financial tightness, highlighting potential limitations of traditional conditional mean estimations and suggesting the value of evaluating tail vulnerabilities. Extending this rationale implies that similar non-linear, tail-heavy dynamics might govern other critical macroeconomic variables that are frequently subjected to sudden, severe structural shocks.

Recent literature has adapted this tail-risk perspective to investigate the effects of geopolitical uncertainty on energy markets. Brignone, Gambetti, and Ricci (2025) examine the non-linear transmission mechanisms of geopolitical risk shocks, suggesting that large positive shocks can trigger a disproportionate amplification channel driven by rising uncertainty. By decomposing geopolitical risk, they observe that geopolitical acts often function similarly to negative demand shocks, whereas geopolitical threats tend to behave as positive uncertainty shocks that significantly elevate oil prices and inflation expectations.

Similarly, Verduzco-Bustos and Zanetti (2026) utilize high-frequency data to isolate geopolitical oil price shocks, finding that these events typically resemble severe supply disruptions, inducing immediate production declines and sharp price increases. Furthermore, they note that these shocks are coupled with heightened uncertainty that prompts a distinct precautionary inventory response, characterized by an initial short-term decline followed by long-term accumulation.

Complementing these findings, Jia, Lyu, and Zhu (2024) applied quantile regression and Growth-at-Risk models to crude oil futures to capture the heterogeneous and asymmetric effects of geopolitical risk and economic policy uncertainty. Their analysis indicates that the influence of geopolitical risk persists over a longer duration, primarily exacerbating the upside risk of price growth, whereas economic policy uncertainty exerts more pronounced effects in the short to medium term. Collectively, this literature underscores the need for policymakers and market participants to consider the asymmetric and heterogeneous effects of uncertainty measures, utilizing quantile-based frameworks to accurately capture the tail characteristics of crude oil prices under geopolitical duress. By synthesizing the Vulnerable Growth methodology with energy-specific geopolitical indicators, the current study aims to bridge a critical gap in the literature. This integration allows for the construction of dynamic risk profiles that move beyond static average estimates, ultimately offering a more comprehensive analytical tool for identifying and hedging against extreme market vulnerabilities.

3 Contribution to literature

This thesis makes six contributions to the literature on commodity market risk and econometric methodology.

- First, it introduces the concept of Oil-at-Risk (OaR), adapting the Growth-at-Risk (GaR) framework of Adrian, Boyarchenko, and Giannone (2019) to the crude oil market. This constitutes the first systematic application of the GaR paradigm to energy market vulnerabilities, enabling probabilistic tail-risk assessments of oil prices in the same language used for macrofinancial surveillance.
- Second, it integrates Conditional Autoregressive Value-at-Risk (CAViaR) models (Engle and Manganelli, 2004) into the OaR framework. This enrichment replaces the static quantile regression step of the canonical GaR procedure with a dynamic quantile process, capturing the persistence and asymmetric adjustment of tail risk that are empirically prominent in oil price behaviour.
- Third, it departs from the conventional use of the skewed-t distribution for distributional fitting by adopting the Johnson SU (JSU) family (Johnson, 1949). The JSU system provides a four-parameter characterisation capable of matching arbitrary skewness and kurtosis combinations, and a formal comparison against the skewed-t benchmark constitutes a novel distributional contribution to the energy-finance literature.
- Fourth, it constructs an entropy-based early warning system (EWS) that exploits highfrequency price data to generate timely signals of oil market stress. Shannon and permutation entropy measures are calibrated against the OaR threshold, providing a model-free leading indicator of vulnerability buildup that is detectable before the lowfrequency distributional model registers elevated downside risk.
- Fifth, it synthesises five heterogeneous bodies of work, namely the GaR and systemic risk literature, quantile regression econometrics, extreme value theory, energy economics, and information theory, into a unified and rigorous testing framework. A comprehensive battery of backtesting diagnostics, including Kupiec unconditional coverage tests, Christoffersen conditional coverage tests, and dynamic quantile tests, is applied uniformly across all model variants, filling a gap in the commodity risk literature where distributional assumptions are seldom subjected to systematic scrutiny.
- Sixth, it delivers a self-contained open-source Python module that implements the full OaR pipeline, covering data preprocessing, quantile regression estimation under both OLS and CAViaR specifications, distributional fitting, EWS computation, and backtesting diagnostics. This research infrastructure contribution promotes replicability and lowers the barrier for future researchers seeking to apply or extend the framework.

4 Data Description

The primary dependent variable is the European Brent crude oil spot price, measured in U.S. dollars per barrel and sourced from the U.S. Energy Information Administration (via FRED). The daily sample begins on Jan 02, 1990 and ends on Apr 30, 2026 providing a substantial historical window that captures numerous major geopolitical crises. Brent is often preferred as a global benchmark because it accounts for approximately two-thirds of the globally traded crude oil supply and tends to reflect international seaborne market conditions more accurately. To address non-stationarity, raw prices are transformed into daily log returns, and weekends and recognized market holidays are removed to maintain a continuous trading day alignment.

The main explanatory variable is the daily Geopolitical Risk Index (GPRD), an automated, dictionary-based measure that calculates the share of newspaper articles discussing adverse geopolitical events and tensions. The index is augmented with temporal lags to model the delayed integration of news into spot market pricing, and country-specific versions may be utilized to evaluate localized risk origins.

To help ensure that the quantile regression accurately isolates the specific effect of geopolitical risk, a targeted suite of exogenous control variables is incorporated into the model:

- **U.S. Dollar Index:** Tracks the relative strength of the United States dollar against a basket of major foreign currencies. Because Brent crude is globally priced in dollars, an appreciation of the currency tends to suppress global demand and lower the commodity's nominal price. Including this variable helps prevent macroeconomic exchange-rate dynamics from being conflated with geopolitical shocks. The data is expressed as a log-difference to enforce stationarity and capture immediate exchange-rate elasticity.
- **Realized Volatility:** Measures the magnitude of recent price dislocations to explicitly control for volatility clusters and latent heteroskedasticity. By utilizing the lagged historical variations of crude returns, this metric serves as a sensitive barometer for the baseline level of underlying market risk prior to a geopolitical event. Controlling for realized volatility helps purge the residual variance of financial panic, isolating the structural pricing of the geopolitical threat itself.
- **Baltic Dry Index (BDI):** A daily composite index measuring the average cost of transporting major raw materials by sea. The BDI serves as a real-time proxy for global industrial demand and real economic activity. Because it tracks physical shipping constraints rather than financial derivatives, it is largely devoid of speculative noise. Including the BDI controls for fundamental demand-side pressures, ensuring the model better isolates true geopolitical shocks from standard macroeconomic business cycles. The daily values are transformed into log-differences to capture the high-frequency growth rate of global shipping demand, and correct for heteroskedasticity.

5 Methodology

When examining the effect of Geopolitical Risk on the rate of change of Brent crude prices, traditional linear methodologies immediately reveal their limitations. An initial look at the data displays a high presence of influential observations, data points located far outside the dense central cloud that heavily distort Ordinary Least Squares regressions. Because OLS minimizes the sum of squared residuals, it is highly sensitive to these extreme outliers, and the resulting slope coefficients often collapse close to zero, offering virtually no predictive power. Linear methods focus entirely on the expected value or conditional mean, effectively ignoring the most critical and overlooked parts of the distribution: the tails.

This empirical challenge aligns with an extensive and ongoing literature in energy econometrics regarding the potential non-linearities of the oil market. As argued by Hamilton (2003), modeling oil shocks often benefits from a flexible approach to account for the possibility that price increases may have a vastly different macroeconomic impact than price decreases. Kilian and Vigfusson (2011) provided compelling evidence suggesting that the macroeconomic response to energy shocks can be asymmetric. Consequently, deploying a strictly linear methodology risks underestimating downside vulnerabilities. The exposure of the market might not be adequately captured by median forecasts; it typically requires a framework specifically equipped to model asymmetric impacts.

Geopolitical shocks introduce an additional layer of non-linear complexity. Pinchetti (2024) suggests that large geopolitical shocks significantly increase Brent prices and inflation expectations by triggering precautionary wait-and-see behaviors and a surge in speculative demand for crude. This is further contextualized by Blanchard and Galí (2010), who highlight how modern oil shocks interact differently with the macroeconomy than they did in the 1970s, mediated by factors such as wage rigidities, central bank credibility, and the shifting energy intensity of the economy. Crucially, the physical materialization of a threat is not required to damage the economy. Kilian, Plante, and Richter (2026) find that an unanticipated increase in the probability of a major geopolitical oil production shortfall, a perceived disaster scenario, can be sufficient to generate a massive surge in oil prices and macroeconomic uncertainty. The Growth-at-Risk methodology, adapted here as Oil-at-Risk, provides an ideal econometric solution to capture this dynamic. By isolating specific quantiles, it explicitly measures how surges in geopolitical uncertainty fatten the tails of the distribution, allowing for a highly nuanced analysis of vulnerabilities that linear models frequently miss.

To quantify these tail risks and vulnerabilities conditional on changes in geopolitical and macroeconomic regressors, the empirical procedure is structured in sequential steps. Rather than focusing solely on the conditional mean, the procedure utilizes the linear quantile regression methodology introduced by Koenker and Bassett (1978) to estimate the conditional quantile function of the future variable. The regression slope for each targeted quantile is obtained by minimizing an asymmetrically weighted sum of absolute errors. By altering the weighting parameter, which ranges continuously between 0 and 1, the regression unevenly distributes the estimation error to place higher weight on specific tail events. In practice, this step estimates a targeted set of percentiles (typically the 5%, 25%, 50%, 75%, and 95% quantiles), tracing out the conditional distribution to evaluate

how Geopolitical Risk uniquely influences both the median and the extremes.

The raw output from the quantile regressions provides a discrete inverse cumulative distribution function. However, due to estimation noise and approximation errors inherent in high-frequency financial data, this discrete output is difficult to map directly into a probability density function. To resolve this, the empirical quantiles are smoothed by fitting a highly flexible continuous distribution. While the original framework by Adrian, Boyarchenko, and Giannone (2019) uses the Azzalini skewed t-distribution, modeling a commodity return may benefit from a more finance-friendly alternative such as Johnson’s S^U distribution. Highly flexible distributions utilize four parameters to shape the curve: the location parameter, the scale or dispersion parameter, the degrees of freedom governing the fatness of the tails, and the shape parameter governing skewness. For each time period in the sample, these four parameters are pinned down by minimizing the squared distance between the empirically estimated quantiles and the theoretical quantiles of the chosen distribution, transforming the discrete points into a continuous density forecast.

Once the continuous conditional distribution is built, the final step evaluates the precise magnitude of the upside and downside risks, revealing how vulnerable the baseline forecast might be to unexpected shocks. This vulnerability is primarily measured using two metrics. First, the structural shift in risk is quantified using Kullback-Leibler (KL) Divergence (often referred to as relative entropy in information theory). KL Divergence serves as a flexible, generalized metric that calculates how the newly estimated conditional probability density diverges from a chosen baseline reference. By evaluating this divergence specifically within the tails of the distribution, the metric captures the informational shift, or “extra surprise,” indicating how much the probability of extreme left-tail or right-tail outcomes has structurally deviated from historical expectations under the prevailing geopolitical conditions. Second, Expected Shortfall and Expected Longrise summarize the tail behavior in terms of absolute severity for a chosen target probability, such as 5%. Expected Shortfall measures the *average expected loss* conditional on the return falling beyond the Value-at-Risk threshold (representing severe price collapses), while the Expected Longrise measures the *average expected gain* conditional on the return falling into the extreme upper tail (representing severe price spikes), finalizing the Oil-at-Risk metric.

5.1 Some theory on Quantile Regression

While standard Ordinary Least Squares (OLS) regression provides a powerful tool for estimating the Conditional Expectation Function (predicting the average value of y given X), it operates under strict assumptions: it is highly sensitive to outliers, requires sufficient normality in the error terms, and fundamentally ignores the tails of the data distribution. Quantile Regression bypasses these limitations by estimating the Conditional Quantile Function (CQF). Instead of modeling the mean, it models a specific quantile τ (e.g., the 10th or 90th percentile), making it distribution-agnostic and robust to extreme outliers. The CQF, given a vector of regressors X_i , is the inverse of the conditional cumulative distribution:

$$Q_\tau(Y_i|X_i) = F_Y^{-1}(\tau|X_i)$$

Just as OLS derives the conditional mean by minimizing the expected squared error, Quantile Regression derives the CQF by minimizing an expected asymmetric loss:

$$Q_\tau(Y_i | X_i) = \arg \min_{q(X_i)} E [\rho_\tau (Y_i - q(X_i))]$$

Where, $\rho_\tau(u)$ is the "Check Function,"

To trace out specific percentiles of a distribution, the regression must penalize under-predictions and over-predictions asymmetrically. If we are estimating the 90th quantile ($\tau = 0.90$), the model should heavily punish under-predictions (negative residuals) and lightly punish over-predictions (positive residuals). To achieve this while maintaining the convexity required for optimization, we construct a piecewise linear function based on the absolute error.

$$|u_i| = \begin{cases} u_i & \text{if } u_i \geq 0 \\ -u_i & \text{if } u_i < 0 \end{cases}$$

By multiplying the positive piece by the target quantile τ and the negative piece by $(1 - \tau)$, we introduce the necessary asymmetry. The Check Function is formally written as:

$$\rho_\tau(u_i) = \begin{cases} \tau u_i & \text{if } u_i \geq 0 \\ (1 - \tau)(-u_i) & \text{if } u_i < 0 \end{cases}$$

This can be written using an indicator function $1_{(u < 0)}$ that takes the value of 1 if the residual is negative and 0 otherwise:

$$\rho_\tau(u) = u \cdot (\tau - 1_{u < 0})$$

Defining the residual as $u_t = y_t - x_t\beta_\tau$, our ultimate objective is to find the parameter vector $\hat{\beta}_\tau$ that minimizes the sum of this asymmetric check function across all observed periods:

$$\hat{\beta}_\tau = \arg \min_{\beta_\tau \in \mathbb{R}^k} \sum_{t=1}^T (\tau \cdot 1_{(y_t \geq x_t\beta_\tau)} |y_t - x_t\beta_\tau| + (1 - \tau) \cdot 1_{(y_t < x_t\beta_\tau)} |y_t - x_t\beta_\tau|)$$

Check the appendix for more details on the optimization procedure.

5.2 Empirical Specification

The empirical specification retained for in-sample estimation regresses daily Brent crude oil log-returns on a seven-day moving average of the Geopolitical Risk index (GPRD-MA_{7,t}) and a vector of control variables $\mathbf{X}_t$.

For each quantile in the target set $\tau \in \{0.01, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95, 0.99\}$, the empirical equation is:

$$Q_\tau(\Delta Brent_t) = \beta_0(\tau) + \beta_1(\tau) \text{GPRD-MA}_{7,t} + \boldsymbol{\gamma}(\tau)' \mathbf{X}_t, \quad (1)$$

where the control vector is:

$$\mathbf{X}_t = (\Delta \text{USDI}_t, \text{RV}_{t-1}, \Delta \text{BADI}_t)'. \quad (2)$$

A single specification is imposed uniformly across all quantiles to preserve comparability across the conditional distribution and to prevent tail-specific overfitting.

The primary regressor is a seven-day moving average of the Geopolitical Risk index:

$$\text{GPRD-MA}_{7,t} = \frac{1}{7} \sum_{j=0}^6 \text{GPRD}_{t-j}, \quad (3)$$

where the weekly window is calibrated to the risk-transmission mechanism operating through financial markets. Geopolitical shocks rarely resolve within a single trading session; their propagation from the news cycle to market positioning typically unfolds over several days as risk premia adjust, hedging demand builds, and institutional investors reassess exposure. Smoothing over seven days therefore captures the sustained escalation of international tensions while filtering the transient spikes generated by individual headlines, ensuring that $\hat{\beta}_1(\tau)$ reflects structural geopolitical stress rather than short-lived media events.

Realized volatility enters the regression at lag $t - 1$ rather than contemporaneously, as a deliberate precaution against look-ahead bias. Including RV_t would introduce information leakage: because realised volatility at time t is computed from intraperiod price movements, it is partly determined by the dependent variable itself, inducing a look-ahead bias. Using RV_{t-1} ensures the regressor is fully observable before period t opens and contains no information unavailable to a market participant at the start of the period.

Collectively, $\text{GPRD-MA}_{7,t}$ and ΔBADI_t decompose geopolitical risk transmission into two complementary channels: the former captures the expectational component embedded in news flow, while the latter captures the physical disruption channel through realised freight costs, as discussed in previous sections.

Equation (1) represents the parsimonious specification selected after screening a wider set of candidate regressors. Among the alternatives evaluated but not retained were measures of real economic activity (industrial production indices and PMI composites), weekly U.S. crude oil supply and production data from the Energy Information Administration, and the Henry Hub natural gas spot price. These variables were excluded on grounds of multicollinearity with the retained controls, insufficient within-sample variation at the daily frequency, or failure to improve out-of-sample quantile score performance.

6 Diagnosis of the Quantile Regression

The diagnostic battery described in previous sections provides broad empirical support for the validity of specification (1). The Koenker-Machado $R^1(\tau)$ confirms that the inclusion of the regressors yields a meaningful reduction in the weighted objective function relative to the intercept-only benchmark, with explanatory power concentrated in the lower tail quantiles, precisely where the Oil-at-Risk metric is most informative. The QAR unit root test rejects the null of quantile-specific non-stationarity at all targeted quantiles, confirming that Brent log-returns exhibit mean-reverting behaviour throughout the conditional distribution. The ARCH-LM test fails to reject the null of no residual autoregressive conditional heteroskedasticity at conventional significance levels, indicating that the specification has successfully absorbed the volatility clustering inherent in daily crude oil returns. The Wald test for equality of slopes across quantiles rejects the null of coefficient constancy, formally confirming that the regressors exert heterogeneous effects across the conditional distribution, which constitutes the foundational identifying assumption of the Oil-at-Risk framework. No quantile crossing violations are detected in the fitted paths, confirming that the estimated conditional distribution satisfies the monotonicity requirement across all observations in the sample. Full tabulations of all test statistics, degrees of freedom, and associated p-values are reported in Appendix ??.

Among these diagnostics, the Dynamic Quantile test of Engle and Manganelli (2004) warrants particular attention, as it is the most direct and demanding test of conditional quantile model validity in a risk management context.

6.1 Dynamic Quantile Test

To verify that the estimated specification correctly captures the time-series structure of the dependent variable, the sequence of quantile violations must be unpredictable given past information. The Dynamic Quantile test of Engle and Manganelli (2004) formalises this requirement by defining the centred hit sequence:

$$Hit_t(\tau) = \begin{cases} 1 - \tau & \text{if } y_t < \hat{q}_t(\tau), \\ -\tau & \text{if } y_t \geq \hat{q}_t(\tau), \end{cases} \quad (4)$$

which has mean zero by construction. If the model correctly specifies the conditional quantile dynamics, violations must be serially uncorrelated and independent of all information available at $t - 1$. The null hypothesis is therefore:

$$H_0 : E[Hit_t(\tau) \mid \Omega_{t-1}] = 0. \quad (5)$$

To test this, the hits are regressed via OLS on an instrument matrix $\mathbf{Z}_t$ comprising a constant, lagged values of $Hit_t(\tau)$, and lagged fitted quantiles. The test statistic,

$$DQ = N \cdot R^2 \xrightarrow{d} \chi_q^2, \quad (6)$$

is asymptotically distributed as χ^2 with q degrees of freedom, where q is the number of instruments in $\mathbf{Z}_t$. Rejection of the null indicates residual predictability in the hit sequence, signalling that the specification fails to absorb relevant tail dynamics.